Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/16/2025
PI Name:	Panontin, Tina Ph.D.		
Project Title:	Content and Representation of Information Needed to Support Time-Constrained Problem Solving		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	tina.panontin@sjsu.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	650.604.6757
Organization Name:	San Jose State University Research Foundation Inc		
PI Address 1:	One Washington Square		
PI Address 2:			
PI Web Page:			
City:	San Jose	**State:**	CA
Zip Code:	95192	**Congressional District:**	18
Comments:	tina.panontin@sjsu.edu; tina.l.panontin@nasa.gov		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/13/2023	**End Date:**	09/13/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The purpose of this work is to assess the content and representation of information needed to support time-constrained problem solving and decision-making during procedure execution and anomaly resolution by the crew under conditions of reduced and/or intermittent communication with the ground team. Research has shown that the specific information accessed and assembled for problem solving typically includes declarative (e.g., facts), procedural (e.g., rules), schematic (e.g., principles), and strategic (e.g., heuristics) types. Further, successful problem solving requires organization and integration of this information into appropriate mental models that can then be exercised to produce solutions. The focus of this task will be on how onboard data systems can enable or promote such models.		

Rationale for HRP Directed Research:	This proposal qualifies for directed study due to the constrained nature of the work. To perform the proposed work, investigators will need to be able to directly communicate with International Space Station (ISS) Flight Controllers, and observe Mission Control Center activities. External researchers would not be able to easily access these personnel and facilities.
Research Impact/Earth Benefits:	
Task Progress:	Historical analyses reveal that, despite the best engineering practices and efforts, significant and persistent rates of anomalies and alarms occur during crewed spaceflight missions. Crews must become more independent from Earth due to the communication latencies of missions beyond low Earth orbit (BLEO). They will need onboard systems to enable urgent problem-solving and support decision-making when the extensive expertise and information resources of the ground team are unavailable. To achieve this, these onboard systems must embody knowledge organization approaches that efficiently promote pattern recognition, explanation, and safe responses by the small, generalist BLEO crew. Using literature reviews, case studies, interviews, and problem-solving process modeling, this task identified essential information to move onboard, and investigated how expert problem solvers organize and integrate such information to illuminate and refine the knowledge and insights contained within. This formed the basis for the human-centered design of a set of data views and information representations utilizing actual International Space Station (ISS) anomaly resolution data and analyses that serve as examples for future testing and development. All the proposed deliverables, as described below, were met. Deliverables: • Annotated literature search on the information needed/used for problem-solving. • Common themes and features of problem-solving models used in time-constrained contexts. • Conceptualization of information content and representation incorporating these problem-solving themes/features and informed by ISS data model. • Findings integrated with deliverables from Mapping BLEO Task (i.e., beyond LEO Human Systems Integration Architecture/HSIA anomaly response – specifying crew roles and responsibilities, teaming, and interactions with onboard systems and delayed ground support). • Countermeasure concepts and requirements to support time-constrained, crew-reliant problem-solving and decision-making during procedure execution and anomaly resolution.
Bibliography Type:	Description: (Last Updated: 01/29/2025)